

1. Data Acquisition
DataBento MBP-10
6 symbols × 4 days
~7M tick events / symbol

2. Offline Feature Research
IC / ICIR scoring per feature
Greedy conditional IC selection
Outputs selected_features.yaml

⋮
optional: reduced
feature config

3. Data Preparation
FeaturePipeline.fit on train only
Welford online z-score normalisation
18 memmap files (6 symbols × 3 days)
val + test DataFrames

4. Config Guardrails
50 pre-flight checks
FATAL → ValueError / abort
WARN → prompt or auto-proceed

pass

5. RL Training
Algorithms: TD3 · DDPG · PPO · Random
StreamingTradingEnvXY episode shuffle
DSR or log-return reward
Runtime health monitor (early stop)
Checkpoints + MLflow logging

6. Post-Training Evaluation
Deterministic rollout on val + test
Sharpe · Sortino · drawdown metrics
Buy-and-hold / TWAP / VWAP benchmarks
Statistical significance tests
Permutation + IG explainability